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3.3.2 Householder transformation

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transformation
that mirrors. Okay?
Well let's see how we
can come up with
that. If we make this

into a rectangle, then
we know that this is
the same as that.

Notice that this right
here is the dot
product of u with x
times the vector u .

It's the component of
 x in the direction of u .
And therefore this
right here from

this point right here is
also $u \cdot x$ times u . And
therefore

this vector, but
pointed in the
opposite direction, is

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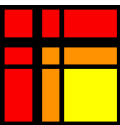
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Reading assignment

1/1 point (graded)



Read Unit 3.3.2 of the notes. [\[LINK\]](#)

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1/1 point (graded)
Show that if H is a reflector, then

- $HH = I$ (reflecting the reflection of a vector results in the original vector).
- $H = H^H$.
- $H^H H = HH^H = I$ (a reflector is unitary).

☒ done



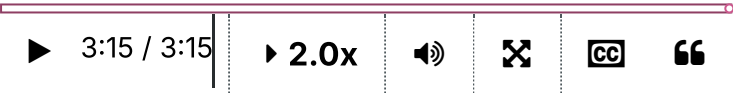
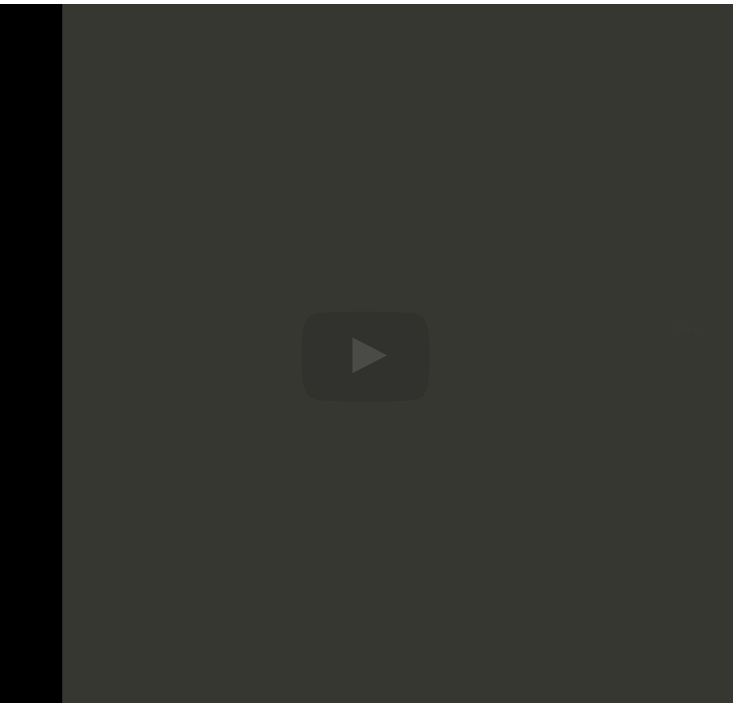
Solution

For a complete solution, see [the notes](#).

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Answers are displayed within the problem

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if you draw a line between the end of vector y and the end of vector x, then the way mirrors work, this is supposed to be perpendicular. So if you can figure out this vector right here, then we know at least the direction of our vector u. Hmm. Let's call that vector v. And what do we know? We know that v plus y gives us x, because the net effect of going here first and then going there is the vector x. And therefore we can see that v is

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approach was to say, "We first compute what this vector v is from here to here." And that vector we knew was equal to x minus y but now y is equal to beta times e_0. Okay and then



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our vector u , which is our joystick, is just v divided by the length of v . Alright, now one possibility would be to say, " Okay, our vector v is equal to -- now let's write our vector x as its individual entries, chi_0 , chi_1 , through chi_{n-1} , and notice that we want to subtract off -- well



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